

E-Verb

Four European broadcast reverbs of the 1980s in one, each with its own faceplate.



Overview

E-Verb recreates four styles of European broadcast reverb from the 80s, built for acoustic precision rather than lushness: there is no chorusing in the tail, and the input runs through a transformer. The MODEL keys change both the algorithm and the faceplate.

ROOM SIM works out the early reflections from the shape and size of a real room, then feeds an unmodulated network that holds up in mono. BROADCAST is a compact, warm plate and hall. DIALOGUE is a dense, quick-recovering reverb tuned for speech. NON-LIN is a gated / reverse processor.

Every parameter sits on its own seven-segment display, with the reverb time set in seconds and adjusted separately at the low and high end, and the room size given in cubic metres.

Reach for it for

- Natural rooms for voices, acoustic instruments and drums
- Speech and voice-over that needs space without blur
- Warm plates and halls with a vintage converter sound
- Gated and reverse effects on drums
- Mono-compatible reverb

Controls

Using the displays

Every parameter has its own display. Click a display to hand it to the data wheel (its frame lights up); drag the display up or down (hold Shift for fine steps), use the mouse wheel over it, or press its - and + keys (hold them to repeat). Double-click a display to return it to its default.

Meters

IN / REV

Two LED columns in 3 dB steps: the input level and the reverb's output level.

MODEL

ROOM SIM

A room simulator: the reflections are worked out from the room's shape (first and second order, with a slight difference between the ears), then an 8-line network with no modulation at all - true room modes that hold up in mono.

BROADCAST

The compact broadcast plate / hall: damped parallel combs into allpasses, a warm loop and a heavier input transformer.

DIALOGUE

A dense allpass tank tuned for speech: high diffusion, fast recovery, no flutter.

NON-LIN

A long tapped line with a shaped envelope instead of a natural decay. TIME sets the length of the shape (up to about 1.25 s, stretched further by SIZE).

SHAPE (RM / GT / RV)

NON-LIN only. RM: a falling, room-like envelope. GT: gate - flat, then cut off. RV: reverse - the sound rises, then stops.

VINTAGE

Runs the reverb through the converters of the era, with the clock rate and bit depth of each model (14-bit on BROADCAST, 16-bit on the others; 31.25 to 40 kHz).

1st REFLECTION

DELAY

When the early reflections start, 0 to 120 ms.

LEVEL

The level of the early reflections, -30 to +6 dB; fully down reads OFF.

REVERB**DELAY**

The pre-delay before the reverb tail, 0 to 250 ms.

LEVEL

The level of the tail, -30 to +6 dB; fully down reads OFF.

REVERB TIME**HIGH END**

A multiplier for the decay time of the highs, 0.1 to 1.5 x. Below 1 the top dies away faster, like real air absorption.

TIME

The decay time at mid frequencies, 0.1 to 20 seconds.

LOW END

A multiplier for the decay time of the lows (below about 250 Hz), 0.5 to 2.5 x.

ROOM**SIZE**

The room volume, 10 to 12000 cubic metres. It sets the spacing of the reflections and the size of the reverb network (and the length of the NON-LIN shape).

REFLECTIONS**DENSITY**

How many early reflections are used, 0 to 100%. Low settings give a few distinct reflections; high settings a dense cluster.

DIFFUSION

How much the input is smeared before it enters the reverb, 0 to 100%. More diffusion gives a smoother, less grainy onset.

FILTER**LO CUT**

A high-pass on the reverb, 20 to 1000 Hz.

HI CUT

A low-pass on the reverb, 1 to 20 kHz (shown in kHz).

CHARACTER

DRIVE	The input transformer, 0 to 100%: the lows saturate first, for weight and warmth. BROADCAST's transformer is heavier.
WIDTH	The stereo width of the reverb, 0 to 150%. 0 is mono; above 100 is wider than natural.
DRIFT	0 to 100%. A slow wander of the air absorption (the high-end damping) - never the delay lengths, so there is no pitch movement. On NON-LIN the tap levels also shift slightly. The wander speeds up as DRIFT goes up.

OUTPUT

MIX	Dry / reverb balance, 0 to 100% (30% by default).
------------	---

PARAMETERS

Data wheel	Changes whichever display was clicked last (named under SELECTED). Drag it (Shift for fine steps) or use the mouse wheel.
Program display	Shows the preset in use (P1 to P10), or P-- when none is loaded.
LIST	Opens the list of all presets by name. The last entry resets everything to the defaults.
◀ / ▶	Step to the previous or next preset.
1 - 9, 0	Memory keys: recall presets 1 to 9, and 0 for preset 10.

IN / OUT

TRIM	Input level before anything else, -24 to +12 dB.
GUARD	Off by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS. It never boosts quiet sources.

SAT	Band saturation on the way out, 0 to 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
LEVEL	Output level, -24 to +12 dB.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency) that holds the peaks at -0.3 dBFS, or a lookahead limiter (3 ms latency, compensated by REAPER) that holds them at -0.3 dBFS. CLIPPING shows under the keys while the clipper is working.

The header

Model name	The line beside the name changes with the model: DIGITAL ROOM SIMULATOR, DIGITAL BROADCAST REVERBERATOR, CONSOLE REVERBERATION UNIT or DIGITAL REVERBERATOR / PROCESSOR.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

The presets set the model and every reverb parameter including WIDTH and MIX; DRIFT, LEVEL and the IN / OUT section are left alone.

Vocal Air	ROOM SIM, a 600 m3 room with 1.4 s decay and a slightly wide image, for vocals.
Concert Room	ROOM SIM, a large 4000 m3 room, 2.2 s with longer lows and darker highs, 30 ms pre-delay.
Drum Room	ROOM SIM, a small 180 m3 room, 0.8 s, close dense reflections and more drive.

Broadcast Plate	BROADCAST, 1.8 s with fast-dying highs, high diffusion and quiet early reflections.
Warm Hall (vintage)	BROADCAST, a 3 s hall with longer lows, darker top and VINTAGE on.
Voice-Over Booth	DIALOGUE, a tiny 60 m3 space at 0.5 s with a 200 Hz low cut, narrower image.
Narration Space	DIALOGUE, a 400 m3 room at 1.2 s for narration.
Gated Snare	NON-LIN with the GATE shape, no early reflections, wide.
Reverse Swell	NON-LIN with the REVERSE shape, a long 3.5 s setting and 20 ms pre-delay.
Tight Ambience	ROOM SIM, a 120 m3 room at 0.4 s with the tail turned down - ambience rather than reverb.
Reset all to defaults	Every control back to its starting value (from LIST).

Tips

- For a natural room, start in ROOM SIM, set SIZE to the space you imagine, then adjust TIME; it holds up well when the mix is checked in mono.
- Pull HIGH END below 1 to make a long reverb sound darker and further away without touching HI CUT.
- On speech, DIALOGUE with a short TIME and LO CUT around 150 - 200 Hz adds space without smearing the words.
- For the 80s gated drum sound, use NON-LIN with GT and set TIME to choose how long the gate stays open.

File: gfx-e-verb.jsfx